Tiny Models Learn Fast

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Abstract

We show that a very small neural network can learn a toy task in a handful of epochs. We describe the model, the loss we minimise, and a comparison against two baselines.

1 Introduction

Large models dominate recent work [1, 2], but small models remain useful on devices with little memory (Smith et al., 2020). Prior studies of compact networks [3] focus on image tasks; we study a simple sequence task instead.[1]

2 Method

Given inputs x_i with labels y_i, the model predicts $\hat{y}_i$ and is trained to minimise the cross-entropy loss:

$$L(\theta) = -\frac{1}{N} \sum_{i} y_i \log \hat{y}_i \qquad (1)$$

Here θ are the model parameters and N is the number of training examples. Equation (1) is minimised with stochastic gradient descent.

[1] Code and data are available from the authors on request.

3 Results

Figure 1 shows the training loss. Table 1 compares our model against two baselines.

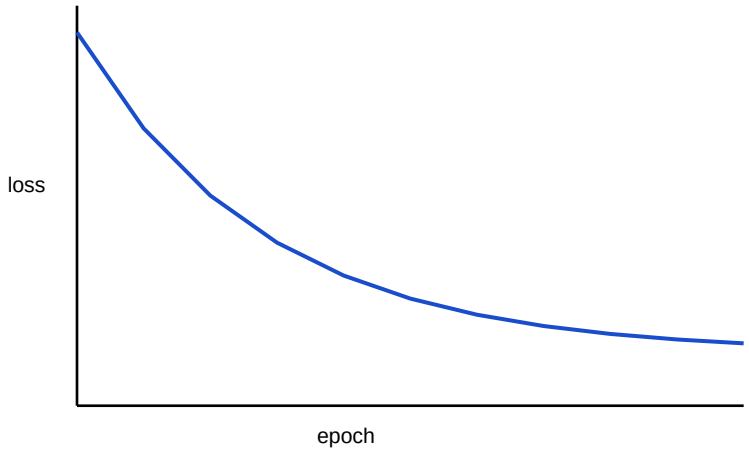


Figure 1: Training loss of the tiny model. Loss falls quickly in the first five epochs and then flattens out.

Table 1: Accuracy on the toy task.

Model	Params	Accuracy
Baseline A	5M	81.2%
Baseline B	12M	84.0%
Ours	0.4M	88.5%

The training loop is short:

```
for epoch in range(10):
    for x, y in loader:
        loss = criterion(model(x), y)
        loss.backward()
        optimizer.step()
```

4 Conclusion

A model with 0.4M parameters beats baselines thirty times its size on this task. Future work will test whether this holds on real data.

References

[1] J. Doe. Scaling laws for everything. In Proc. of Big Conf, pages 1-10, 2023.

[2] K. Lee and M. Park. Bigger is better. arXiv:2401.00001, 2024.

[3] L. Smith, R. Jones, and T. Wu. Compact nets. J. Small Models, 4(2):11-19, 2020.